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Persistence in Variance, Structural Change, and the GARCH Model

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This article examines the persistence of the variance, as measured by the generalized autoregressive conditional heteroskedasticity (GARCH) model, in stock-return data. In particular, we investigate the extent to which persistence in variance may be overstated because of the existence of, and failure to take account of, deterministic structural shifts in the model. Both an analysis of daily stock-return data and a Monte Carlo simulation experiment confirm the hypothesis that GARCH measures of persistence in variance are sensitive to this type of model misspecification.

KEY WORDS: Bootstrap; I-GARCH; Integration in variance; Monte Carlo.

1. INTRODUCTION

Persistence in variance of a random variable, evolving through time, refers to the property of momentum in conditional variance; past volatility explains current volatility. The degree to which conditional variance is persistent, or permanent, in daily stock-return data is an important economic issue. Poterba and Summers (1986) showed that the extent to which stock-return volatility affects stock prices (through a time-varying risk premium) depends critically on the permanence of shocks to variance. More generally, modeling the pricing of contingent claims, such as options, relies on perceptions of how permanent shocks to variance are. A current shock of given magnitude that is expected to be transitory will have a smaller impact on the price of an option with many periods to maturity than a shock that is mostly permanent. Effectively modeling volatility persistence is a key aspect of understanding such issues.

The generalized autoregressive conditional heteroskedasticity (GARCH) model, developed by Engle (1982) and Bollerslev (1986), has proven to be a useful means for empirically capturing the momentum in conditional variance. Under GARCH, shocks to variance persist according to an autoregressive moving average (ARMA) structure of the squared residuals of the process. Much of the recent evidence from financial-market data seems to suggest that persistence in variance, as measured by ARCH models, is quite substantial. This apparent empirical regularity has motivated Engle and Bollerslev (1986) to introduce the integrated-GARCH (I-GARCH) process, in which shocks to variance do not decay over time. Integration in variance is analogous to a unit root in the mean of a stochastic process, an example of which is the random walk.

The contention of this article is that it might be misleading to take current evidence of I-GARCH, or in general evidence of strong persistence, at face value. One potential problem of I-GARCH is that, unlike the random walk in mean for asset prices, it lacks theoretical motivation. A possible explanation is that the serial dependence inherent in GARCH proxies for an exogenous driving variable [see, for instance, Lamoureux and Lastrapes (in press)], which itself is integrated in mean, but this link has not firmly been established. In this article, we investigate the possibility that the appearance of high persistence in variance in daily stock-return data is due to time-varying GARCH parameters. In particular, we allow for deterministic, or structural, shifts in the unconditional variance of the stochastic process. Such shifts, if unaccounted for, may bias upward GARCH estimates of persistence in variance and thus vitiate the usefulness of GARCH in situations in which the degree of permanence is important.

Section 2 briefly discusses the GARCH model and motivates the empirical strategy. Section 3 reports estimates of a generalized variance equation that allows for shifts in the constant term for a sample of 30 common-stock daily returns and a stock index. The results support the notion that misspecification of the GARCH model—that is, not accounting for discrete shifts in unconditional variance—can lead to misinterpretation of estimates of persistence in variance. In Section 4, a simulation experiment is conducted in which known structural shifts are ignored in estimation. The simulation provides some insights into the nature of the estimation bias due to misspecification.

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2. THE GARCH MODEL

Consider the following GARCH model:

$$y_t = x_t\beta + \varepsilon_t, \tag{1}$$

$$(\varepsilon_t \mid \varepsilon_{t-1}, \varepsilon_{t-2}, \ldots) \sim N(0, h_t), \tag{2}$$

and

$$h_t = \omega + \alpha_1\varepsilon_{t-1}^2 + \alpha_2\varepsilon_{t-2}^2 + \cdots + \alpha_p\varepsilon_{t-p}^2 + \gamma_1 h_{t-1} + \cdots + \gamma_q h_{t-q}, \tag{3}$$

where x_t is an exogenous, or predetermined, variable and ε_t is a random error. The GARCH (p, q) model in Equations (1)–(3) allows the conditional variance of the random disturbance to depend linearly on the past behavior of the squared errors. As noted by Sastry Pantula (see Bollerslev 1986, p. 310), this model implies that ε_t^2 is generated by an ARMA process. Although time series methods exist that facilitate the choice of optimal values for p and q (Bollerslev 1986), the GARCH(1, 1) specification has proven to be an adequate representation for most financial time series. We therefore use this specification in the following discussion and empirical work.

To understand the nature of persistence in variance under the GARCH(1, 1) model, note that (3) can be rewritten as follows, for $p = q = 1$:

$$h_t = \omega + \lambda h_{t-1} + \alpha v_{t-1}, \tag{4}$$

where $v_{t-1} \equiv \varepsilon_{t-1}^2 - h_{t-1}$ is a serially uncorrelated innovation, $\lambda \equiv (\alpha + \gamma)$, and the subscripts on the parameters have been dropped for notational convenience. By eliminating h_{t-1} from the right side, we can express the conditional variance as

$$h_t = \sigma^2 + \alpha[v_{t-1} + \lambda v_{t-2} + \lambda^2 v_{t-3} + \lambda^3 v_{t-4} + \cdots], \tag{5}$$

where $\sigma^2 \equiv \omega/[1 - \lambda]$ is the unconditional variance. This expression makes clear the dependence of the persistence of volatility shocks (v_t) on the sum of the GARCH parameters, λ. As this sum approaches 1 from below, the effects of past shocks on current variance become stronger. For $\lambda = 1$ (i.e., the lag polynomial of conditional variance contains a unit root), the process is said to be integrated in variance or I-GARCH (Engle and Bollerslev 1986). Shocks to variance do not decay over time in this case; moreover, the unconditional variance does not exist.

As noted previously, a common finding when the GARCH model is applied to high frequency (e.g., daily) asset-price data is that shocks to variance are strongly persistent; that is, λ is very close to 1. Bollerslev (1988) provided a brief discussion of this literature. [Chou (1988) showed that temporal aggregation of the data reduces the measured persistence in GARCH models.] Typically, the results for daily financial market data are based on very long time series. For example, French, Schwert, and Stambaugh (1987), who obtain a sum of the GARCH parameters to be .997, have over 15,000 observations on daily log stock-price relatives. Lamoureux and Lastrapes (in press) (LL), however, obtain GARCH estimates of persistence for individual stocks that are smaller than might be expected given the previous GARCH literature. For daily returns on 20 individual stocks, GARCH(1, 1) models are estimated over a sample of approximately 300 observations. Averaged over the 20 companies, λ is .728. Note that the LL GARCH parameters are statistically significant in most of the individual cases.

A potential explanation for low GARCH values estimated over small samples was suggested by Diebold (1986, p. 55), who noted that I-GARCH might be due to instability of ω over the sample period—that is, nonstationarity of the unconditional variance. This possibility was confirmed by Lastrapes (1989), who showed that exchange-rate volatility, as measured by ARCH, depends on U.S. monetary policy regimes. Including dummy variables to account for regime changes diminishes the degree of ARCH persistence. Given these results, it seems likely that the LL results are due to the sample size being smaller than other studies; the longer the sample period, the higher the probability that structural shifts will be present. The purpose of this article is to provide confirming evidence using stock-return data that, ignoring simple structural shifts in unconditional volatility (i.e., model misspecification), can lead to the spurious appearance of extremely strong persistence in variance.

An empirical strategy is immediately apparent: Determine times at which shifts in the variance constant ω (implying shifts in unconditional variance) are likely to have occurred, and account for these by estimating a more general variance model:

$$h_t = \omega' + \delta_1 D_{1t} + \cdots + \delta_k D_{kt} + \lambda h_{t-1} + \alpha v_{t-1}, \tag{6}$$

where D_{it} $(i = 1, \ldots, k)$ are dummy variables that correspond to periods over which the GARCH process is stationary. Note that there are $k + 1$ such periods given the form of (6). The empirical test of our hypothesis is to compare the implied persistence of the model using (4)—the restricted specification—to that of the unrestricted specification (6).

The primary difficulty in performing this test lies in determining the timing of structural shifts in unconditional variance, especially for individual stocks. Given this difficulty, we estimate the GARCH(1, 1) model for daily stock returns over a relatively large range of data (4,228 observations), including dummy variables for arbitrarily chosen subsamples. We choose to allow for structural shifts every 302 observations; that is, $k = 13$ mutually independent dummy variables are included in (6). These subsamples are large enough to yield reasonable statistical inference, yet small enough

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to make it likely that structural shifts in ω are captured by the model. They are also of the magnitude of the samples in LL.

The objective is to show that persistence in variance can be overstated if nonstationarity in ω is ignored in estimation. This purpose is served even if a systematic search for the timing of structural shifts is not attempted. Furthermore, the variance model (4) could be generalized further by allowing α and γ to be time dependent. We believe, however, that our point can be forcefully made, and more easily grasped, in the context of a simple structural change. Although this approach is ad hoc, we think it is a fair way to illustrate, using actual data, the potential for misspecification of the GARCH model and how this misspecification affects quantitative measures of volatility persistence.

3. THE GARCH(1, 1) MODEL APPLIED TO STOCK RETURNS

In this section, the general GARCH(1, 1) specification of Equation (6) is compared to the restricted variance of (4) using daily stock-return data. The focus is on the effect of the generalization on measured persistence in variance. The data consist of arithmetic returns (based on daily closing prices) of 30 randomly selected stocks for which complete return data is available from January 1, 1963, to November 13, 1979, on the 1987 version of the daily Center for Research in Security Prices (CRSP) tape. The 30 sample companies and their CUSIP numbers are listed in Table 1. The time series of daily returns of the CRSP value-weighted index of New York Stock Exchange and American Exchange stocks, used by Chou (1988), is also analyzed.

Table 1. Companies Used in Estimating GARCH Models

Number	CUSIP	Name
1	03190510	Amoco Corporation
2	74271810	Procter and Gamble
3	02553710	American Electric Power, Inc.
4	01371610	Alcoa Aluminum Ltd.
5	66577210	Northern States Power Corporation (Minnesota)
6	81238710	Sears Roebuck
7	83237710	Smithkline Beckman
8	84240010	Southern California Edison
9	00205010	American South African Investment Ltd.
10	74740210	Quaker Oats
11	73109510	Polaroid
12	75511110	Raytheon
13	72348410	Pinnacle West Capital Corporation
14	86157210	Stone and Webster, Inc.
15	04247810	Armtek Corporation—Delaware
16	11009710	Bristol-Myers
17	15852510	Champion International Corporation
18	23329210	DPL, Inc.
19	29356710	Enserch Corporation
20	37153210	Genesco Inc.
21	42823610	Hewlett-Packard
22	48517010	Kansas City Southern Industries, Inc.
23	58155610	McKesson Corporation
24	42323610	Helene Corporation
25	46107410	Interstate Power Corporation
26	54626810	Louisiana Land and Exploration
27	71404110	Perkin Elmer
28	74457310	Public Enterprise Group
29	80685710	Schlumberger, Ltd.
30	60803010	Mohasco Corporation

For each company in the sample, we estimate two systems, Equations (1), (2), and (4)—the restricted model—and (1), (2), and (6)—the unrestricted model. The variable y_t is the stock return over day t, and the vector x_t contains a constant and y_{t-1}; the lagged dependent variable accounts for nonsynchronous trading. For observations ranging from $(1 + 302i)$ to $[(1 + i)302]$, D_{it} $(i = 1, \ldots, 13)$ is 1, 0 otherwise. The full sample contains 4,228 observations; thus the dummy variables allow for 14 nonoverlapping subsamples (ω' measures the constant variance term over the first subsample).

A hybrid of the Berndt, Hall, Hall, and Hausman (1974) (BHHH) algorithm and a sequential quadratic programming algorithm (Biegler and Cuthrell 1985) is developed and used to obtain parameter estimates that maximize the likelihood function. The gradient is calculated numerically. Parameter estimates are insensitive to various initial conditions for our sample, making it likely that global maxima are achieved.

The estimation results are reported in Table 2. In terms of the restricted model, the parameter estimates for α and γ are consistent with those in the literature. The model implies strong persistence in variance as measured by λ. The average for this sum over the 30 companies is .978. To provide some perspective on the implied persistence, we also report the half-life (HL) of volatility shocks, which depends only on λ. Half-life measures the period of time (number of days) over which a shock to volatility diminishes to half its original size and is given by $\mathrm{HL} = 1 - [\log 2/\log \lambda]$. For the sample stocks, the half-life for the average sum across the stocks is 32.16 days.

The results for the unrestricted model, which includes shift dummies, support our joint hypothesis that shifts in ω can lead to spuriously measured persistence and that such shifts are not unlikely in stock-return data. For all companies, the sum of the GARCH parameters decreases when the dummies are included. Typically, α increases, but the fall in γ more than offsets this rise in α. The average sum in the general case is .817. Half-life falls dramatically as shifts in ω are accounted for, the average over companies declining to 4.43 days from that implied by the restricted model.

The results for the stock-index series are reported in the final row of the table. Although there is a decline in measured persistence from the restricted to the unrestricted case and a concomitant fall in half-life, this decline is not as large as for the individual stocks. There are two explanations of this result consistent with our hypothesis—(a) relevant shifts for individual stocks cancel in aggregation and (b) the unrestricted model uses inappropriately timed dummy variables.

The decline in persistence-cum-structural shifts is large for many companies, but nothing yet has been

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Table 2. Persistence in Variance for 30 Stocks: Restricted [Eq. (4)] and Unrestricted [Eq. (5)] GARCH(1, 1) Specifications

	Restricted model					Unrestricted model			
Company	α	γ	λ	HL	5% CV	α	γ	λ	HL
1	.087	.884	.971	24.55	.942	.123	.654	.777	3.75
2	.053	.935	.988	58.41	.959	.055	.884	.939	12.01
3	.042	.950	.992	87.30	.963	.071	.814	.885	6.67
4	.050	.922	.972	25.41	.943	.063	.817	.880	6.42
5	.035	.950	.985	46.86	.956	.105	.570	.675	2.76
6	.067	.904	.971	24.55	.942	.084	.785	.869	5.94
7	.063	.917	.980	35.31	.951	.084	.817	.901	7.65
8	.047	.944	.991	77.67	.962	.063	.809	.872	6.06
9	.071	.908	.979	33.66	.950	.086	.811	.897	7.38
10	.034	.963	.997	231.70	.968	.138	.550	.688	2.85
11	.025	.970	.995	139.28	.966	.036	.887	.923	9.65
12	.042	.952	.994	116.18	.965	.050	.905	.955	16.05
13	.050	.944	.994	116.18	.965	.080	.691	.771	3.67
14	.093	.897	.990	69.97	.961	.200	.521	.721	3.12
15	.061	.926	.987	53.97	.958	.126	.664	.790	3.94
16	.076	.862	.938	11.83	.909	.147	.494	.641	2.56
17	.059	.936	.995	139.28	.966	.126	.681	.807	4.23
18	.056	.908	.964	19.91	.935	.098	.489	.587	2.30
19	.047	.945	.992	87.30	.963	.103	.775	.878	6.33
20	.114	.898	1.012	—[a]	.983	.234	.453	.687	2.85
21	.081	.881	.962	18.89	.933	.121	.719	.840	4.98
22	.050	.945	.995	139.28	.966	.078	.809	.887	6.78
23	.025	.967	.992	87.30	.963	.022	.959	.981	37.13
24	.037	.954	.991	77.67	.962	.041	.926	.967	21.66
25	.056	.922	.978	32.16	.949	.119	.601	.720	3.11
26	.062	.934	.996	173.94	.967	.082	.847	.929	10.41
27	.152	.740	.892	7.06	.863	.169	.582	.751	3.42
28	.077	.907	.984	43.97	.955	.120	.664	.784	3.85
29	.073	.899	.972	25.41	.943	.111	.670	.791	3.80
30	.127	.742	.869	5.94	.840	.152	.572	.724	3.15
Mean	.064	.914	.978	32.16[b]		.103	.714	.817	4.43[b]
Index	.082	.907	.989	63.67	.960	.080	.878	.958	17.15

NOTE: HL is the half-life of variance; HL = 1 − [(log 2)/log λ]. The bootstrap 5% critical value is denoted by 5% CV.

[a] Because λ exceeded 1 in this case, HL cannot be interpreted. Note that HL approaches infinity as λ approaches 1, so persistence should be considered large for this company.

[b] The mean HL is obtained by applying the half-life formula to the mean of λ.

said regarding the statistical significance of the decline. The GARCH model has been estimated under the assumption that the conditional distribution of ε_t is normal. Application of the Kiefer–Salmon (1983) joint test for skewness and excess kurtosis to the adjusted residuals ($\varepsilon_t h_t^{-1/2}$) shows that normality is rejected for the sample companies. Although Bollerslev and Wooldridge (1988) proved that the maximum likelihood estimator is consistent in the presence of nonnormality, standard error estimates based on the information matrix at the optimum will be incorrect. In addition, dummy variables do not satisfy the conditions necessary for the estimators to have the usual asymptotic properties. Thus, care must be used in constructing test statistics.

The desired test is the null hypothesis that λ in the restricted model equals λ in the unrestricted model, against the alternative that the latter parameter is less than the former. Under the null, high persistence in variance exists (as given by λ in the restricted model), and there are no discrete, structural shifts. To control for Type I error, we need to characterize the sampling distribution of the estimator of λ in the unrestricted model under the null. Monte Carlo simulation is inappropriate, because the distribution of the errors is unknown. The approach we use is the bootstrap technique (Efron 1979) to obtain the empirical distribution of the estimator.

The test of the null is performed as follows. Five hundred bootstrap samples are drawn from the standardized residuals of the restricted GARCH(1, 1) model for company 16. This company is chosen because the adjusted residuals are highly nonnormal and exhibit no serial dependencies. The bootstrap residuals, which contain the characteristics of the actual distribution, are transformed into a true GARCH(1, 1) with $\lambda = .99$. (This sum was not set to 1 to avoid potential "knife-edge" problems of I-GARCH.) For each of the 500 realizations of the true process, the general GARCH model is estimated and the parameters saved. The 500 estimates of λ define the empirical distribution of the estimator under the null. Details of the bootstrap are reported in Appendix A.

To test the null hypothesis for each stock, we choose the fifth percentile value (.9609) of the empirical distribution of λ estimated from the unrestricted model to construct 5% critical values. The fifth percentile is subtracted from the true value of .99; this deviation (.0291) is in turn subtracted from the sum for each company under the restricted estimation. The results are the 5%

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Table 3. Results of Monte Carlo Simulation

	Unrestricted model		Restricted model		
	Mean	Standard deviation	Mean	Standard deviation	True value
			Environment 1		
β_0	−.162	.224	−.163	.229	−.15
	(.218)	(.009)	(.221)	(.010)	
β_1	.010	.039	.009	.036	.01
	(.038)	(.001)	(.039)	(.002)	
ω'	21.346	8.541	1.615	.674	20.00
	(8.388)	(4.520)	(1.890)	(.537)	
α	.146	.049	.102	.022	.15
	(.050)	(.008)	(.051)	(.008)	
γ	.422	.187	.872	.026	.45
	(.182)	(.090)	(.078)	(.013)	
δ_1	21.601	10.005	—	—	20.00
	(9.128)	(5.278)			
δ_2	−10.629	4.905	—	—	−10.00
	(4.877)	(2.472)			
			Environment 2		
β_0	−.148	.158	−.150	.161	−.15
	(.156)	(.006)	(.160)	(.007)	
β_1	.089	.039	.009	.035	−.01
	(.039)	(.001)	(.040)	(.002)	
ω'	21.909	8.873	.649	.280	20.00
	(8.939)	(3.949)	(.964)	(.289)	
α	.149	.050	.086	.019	.15
	(.049)	(.008)	(.054)	(.009)	
γ	.409	.195	.893	.023	.45
	(.194)	(.082)	(.082)	(.014)	
δ_1	−10.831	5.084	—	—	−10.00
	(5.121)	(2.125)			
δ_2	−16.404	6.885	—	—	−15.00
	(6.950)	(3.067)			

NOTE: Means and standard deviations are calculated across 1,000 simulations. The numbers in parentheses are the summary statistics calculated for the estimated coefficient standard deviations.

critical values and are reported in Table 2. The critical value is robust to nonnormality and avoids problems due to the inclusion of dummy variables in the variance equation.

Consider, for example, company 1, for which the model estimates are similar to the mean parameter estimates; the probability that the unrestricted sum lies below .942, given that the null $\lambda = .971$ is true, is 5%. For each company except numbers 23 and 24, the null hypothesis of high persistence is rejected in favor of the alternative. The null is also rejected for the stock index. These results suggest that the diminished persistence under the general variance model is not due to sampling error or bias.

This inference is strictly correct only for company 16. The bootstrap has not been repeated for each company because of the large amount of computational time required. Exploiting a company that has a highly nonnormal residual distribution, however, provides a conservative test of the null. Thus, these findings are consistent with the hypothesis that accounting for discrete shifts in unconditional variance diminishes measured GARCH persistence and reconcile the results of Lamoureux and Lastrapes (in press) to the other GARCH literature.

4. MONTE CARLO SIMULATION

Because the unrestricted model estimated with the stock-return data includes arbitrary dummy variables, and because of the unknown properties of the residual distributions, in this section we analyze measured persistence in variance in the context of a Monte Carlo experiment. We simulate a GARCH process with relatively low persistence and a small number of deterministic shifts in unconditional variance. The objective is to provide further insights into the nature of the noted misspecification bias by controlling for extraneous factors.

To this end, a normal GARCH(1, 1) process is simulated for two environments. In each, the following parameter values are assumed [see Eqs. (1), (2), and (6)]: $\beta_0 = -.15$, $\beta_1 = .01$, $\omega' = 20.0$, $\alpha = .15$, $\gamma = .45$, and $\varepsilon_0 = 0$. Eight hundred one observations are generated for y_t ($t = 0, \ldots, 800$) in each simulation; $k = 2$, so three regimes of unconditional variance, each of 266 observations, are assumed. The two environments differ according to the values of the dummy parameters δ_1 and δ_2—in environment 1, $\delta_1 = 20.0$ and $\delta_2 = -10.0$; in environment 2, $\delta_1 = -10.0$ and $\delta_2 = -15.0$. The length of the regimes is similar to the spacing of the dummies in Section 3. The magnitude of the dummy coefficients is typically smaller than the estimates of the dummies (not reported) for the stock-return data. Note also that $\lambda = .6$ is also generally smaller than those of the individual stocks for the unrestricted model. We simulate these environments to show that the results of Section 3 are robust to the *direction* of structural shifts.

One thousand simulations are conducted for each environment. For each simulation, the restricted and the correct, unrestricted GARCH models are estimated. Table 3 reports the means and standard deviations across simulations of the estimated parameters, and Appendix B contains other details of the simulation.

For the correctly specified model, the table reveals a slight downward bias in the estimate of γ in each environment, although λ deviates only slightly from the true value of .60. (The true half-life is 2.35 days,

Table A.1. Results of Bootstrap: Mean and Standard Deviation of λ Across Bootstraps, as a Function of the Number of Bootstraps

	Unrestricted model*		Restricted model		
	Mean	Standard deviation	Mean	Standard deviation	True values
β_0	.436	.386	.417	.406	.410
β_1	.094	.018	.094	.019	.095
ω	—	—	15.384	4.621	14.404
α	.093	.013	.096	.014	.095
γ	.883	.014	.893	.014	.895

* To save space, coefficients on dummy variables are not reported.

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Table A.2. Results of Bootstrap: Empirical Density of Unrestricted λ

Percentile	*Value*	*Percentile*	*Value*
Minimum	.940	x_{60}	.978
x_1	.951	x_{70}	.980
x_2	.957	x_{80}	.983
x_5	.961	x_{90}	.986
x_{10}	.965	x_{95}	.990
x_{20}	.968	x_{98}	.993
x_{30}	.972	x_{99}	.994
x_{40}	.974	Maximum	.999
x_{50}	.976		

whereas the half-life estimated in environment 2 is 2.19 days.) This result is consistent with the experiments of Bollerslev and Wooldridge (1988) and Engle, Hendry, and Trumble (1985). Note also that the mean across simulations of the (asymptotic) standard error of each parameter estimator (shown in parentheses) is almost identical to the standard deviation of the estimator across simulations.

The simulation result of primary interest is the effect on measured persistence in variance when the misspecified GARCH model is estimated. For environments 1 and 2, respectively, λ is .974 and .979, a clear upward bias in measuring the persistence of volatility shocks. This bias does not appear to be sensitive to the direction of the shifts in the unconditional variance.

The simulations support our interpretation of the stock-return results of Section 3. They also provide insight into the nature of the misspecification bias. Consider a one-time increase in unconditional variance due to an increase in ω. Since low-magnitude shocks cluster

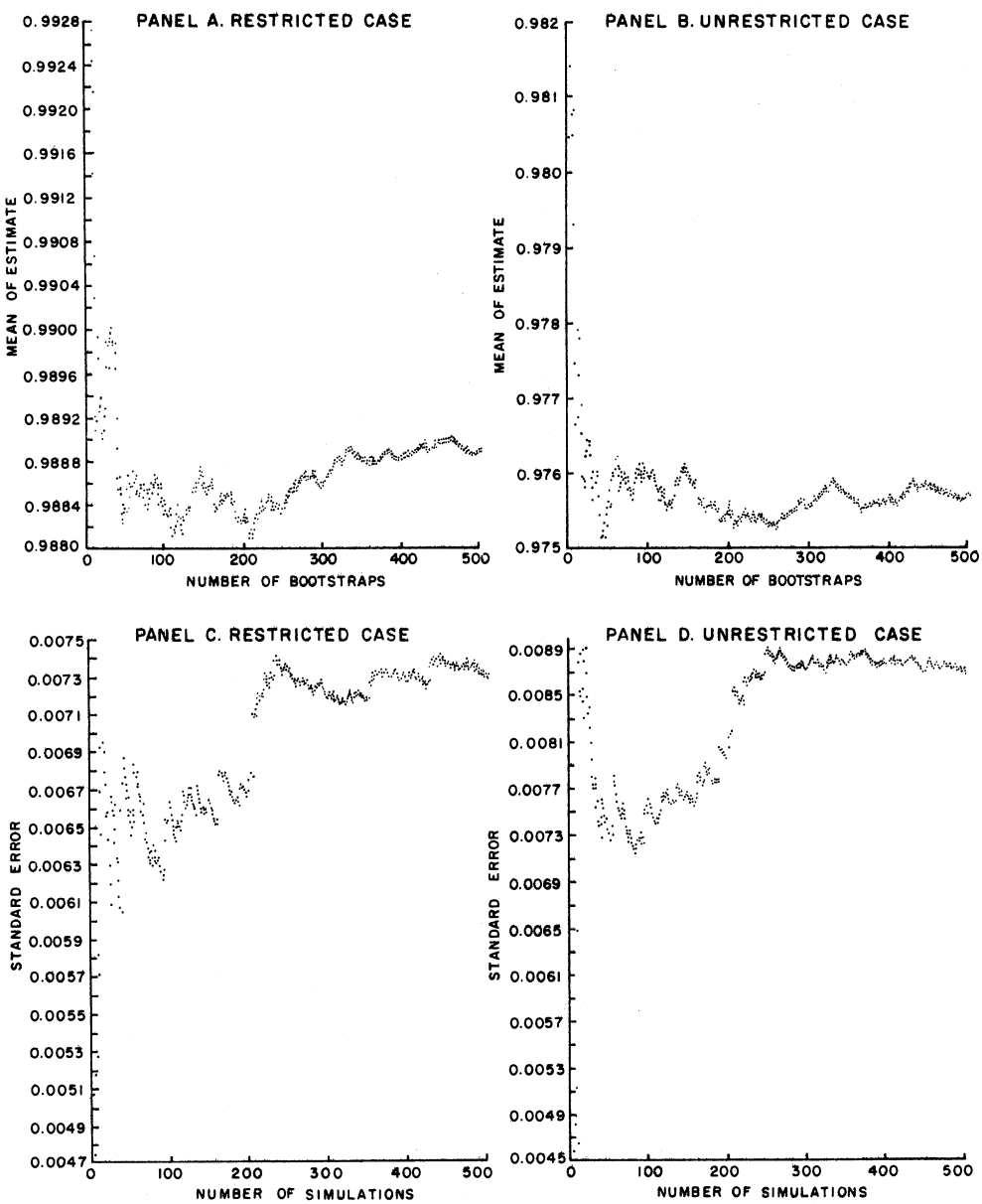


Figure A.1. Means and Standard Deviations of $\lambda = \alpha + \gamma$. Panels A and B report the sample mean of λ across bootstraps as a function of the number of bootstraps for the restricted and unrestricted models, respectively. Panels C and D report the corresponding sample standard deviations. Estimates of λ are obtained from the unrestricted model, given the simulated system in Appendix A.

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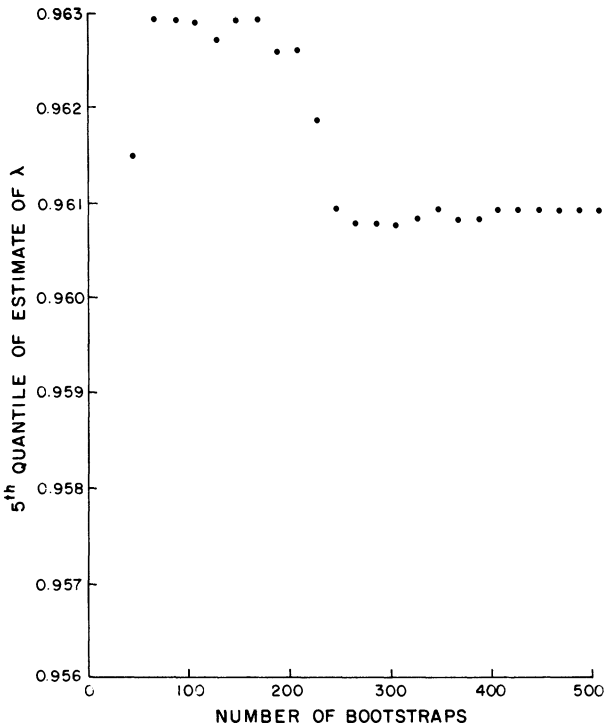


Figure A.2. Bootstrap 5% Quantile of $\lambda = \alpha + \gamma$. The plot shows the 5% quantile of the empirical distribution of λ as a function of the number of bootstrap replications. Estimates of λ are obtained from the unrestricted model, given the simulated system in Appendix A.

in the initial period and high-magnitude shocks cluster subsequent to the shift, the variance process will appear to have momentum; a GARCH model that does not account for the structural change will pick up high persistence. But note that the same will occur if the constant term *decreases* (as in environment 2), since clustering will still be present. The direction of the shift is irrelevant. Therefore, a GARCH model that is misspecified along the lines discussed here will overstate the persistence of shocks to variance.

5. CONCLUSIONS

The maintained hypothesis of this article is that the application of GARCH to long time series of stock-return data will yield a high measure of persistence (close to an I-GARCH process) because of the presence of deterministic shifts in the unconditional variance and the subsequent failure of the econometrician to model these shifts. The results of the dual experimental methodology are consistent with this hypothesis.

Our results suggest at least two directions for future research. First, given that structural shifts are likely to occur for daily stock returns, the next logical step is to develop means of identifying the timing of such discrete shifts. Hamilton's (1988, 1989) method of estimating nonstationary time series may prove to be productive in this regard.

Second, discrete, infrequent shifts in unconditional variance are a type of persistence in variance. The distinction between this type of persistence and GARCH is that under the latter process a given realization contains information about the future evolution, whereas under the former such a shock contains no information. The important question is the role this distinction plays in economic contexts, such as the risk-premium analysis of Poterba and Summers (1986) or models of option pricing.

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APPENDIX A: BOOTSTRAP SIMULATION

The basic sample for the bootstrap is the set of adjusted residuals from the GARCH(1, 1) model [Eqs. (1), (2), and (4)] of company 16, Bristol-Myers. Five hundred random samples of 4,227 observations each are drawn from the adjusted residuals, $\varepsilon_t h_t^{-1/2}$, with replacement using the International Mathematical and Statistical Libraries (IMSL) subroutine GGUD. The adjusted residuals are appropriate for the bootstrap because under the null hypothesis they have been purged of their time series properties. (A Box–Ljung portmanteau test for serial dependence indicates that these residuals are uncorrelated.) Each bootstrap sample is then converted to the following GARCH(1, 1) process:

$$y_t = .409933 + .095\, y_{t-1} + \varepsilon_t, \qquad \text{(A.1)}$$

$$\varepsilon_t \sim F(0, h_t), \qquad \text{(A.2)}$$

and

$$h_t = 14.4038 + .095\, \varepsilon_{t-1}^2 + .895\, h_{t-1}, \qquad \text{(A.3)}$$

where ε_t are the bootstrapped variates, y_0 and ε_0 are specified to be 0, and h_0 is the unconditional variance of the bootstrap sample of ε. The distribution F is not specified but has finite first and second moments by construction. The parameter values in (A.1) and the constant term in (A.3) are the restricted estimates from company 16. The values of the conditional-variance slope parameters are chosen so that $\lambda = .99$.

Each of the 500 generated samples of y_t are estimated with both the restricted and unrestricted specifications of the model. The unrestricted model includes 13 evenly spread dummy variables. The BHHH, Biegler–Cuthrell (1985) hybrid algorithm is used in estimation, since boundary conditions are easy to impose. All parameters in the variance equation are bounded from below by 0 to avoid algorithmic failure for an iteration in which variance is negative. After some experimentation, we verified that the hybrid and BHHH algorithms yield identical results for comparably chosen tolerance levels. All computations for the bootstrap use an IBM 3090-

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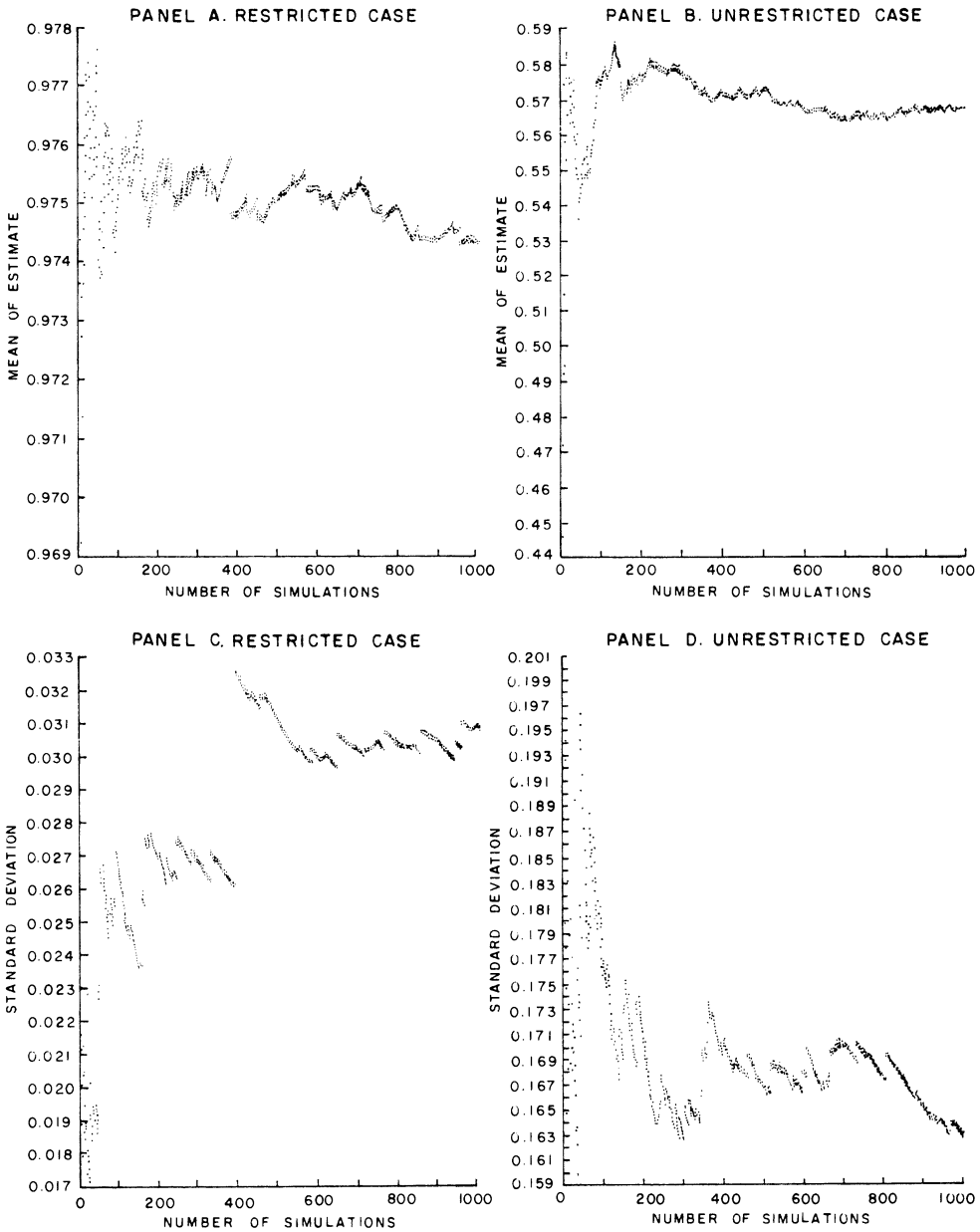


Figure B.1. Means and Standard Deviations of $\lambda = \alpha + \gamma$ Across Monte Carlo Simulations—Environment 1. Panels A and B report the sample means, and Panels C and D report sample standard deviations across simulations as a function of the number of simulations for the restricted and unrestricted models.

600E supercomputer with double-precision FORTRAN; 600 minutes of central processing unit (CPU) time are used. Table A.1 reports the means and standard deviations of the parameter estimates across the 500 bootstrap samples for both models. Table A.2 reports a list of the percentiles of the empirical distribution of λ estimated under the unrestricted model, which is used as discussed in the text.

Figures A.1 and A.2 verify the appropriateness of stopping at 500 bootstrap samples. Figure A.1 plots the means and standard deviations of λ as a function of the number of bootstrap samples. Figure A.2 plots the fifth percentile value of the empirical distribution of λ as a function of the number of bootstrap samples. In each figure, the statistic converges before 400 samples, indicating little gain to extending beyond 500.

We provide no evidence on the robustness of inference with respect to variations in the parameter values of (A.1) to (A.3). Because of the computational expense involved in performing the bootstrap simulation, this issue is left for future research.

APPENDIX B: MONTE CARLO SIMULATION

The Monte Carlo experiment simulates a normal GARCH(1, 1) process with discrete jumps in the constant term in the variance equation. Two environments are simulated, each defined by the type of shift in ω.

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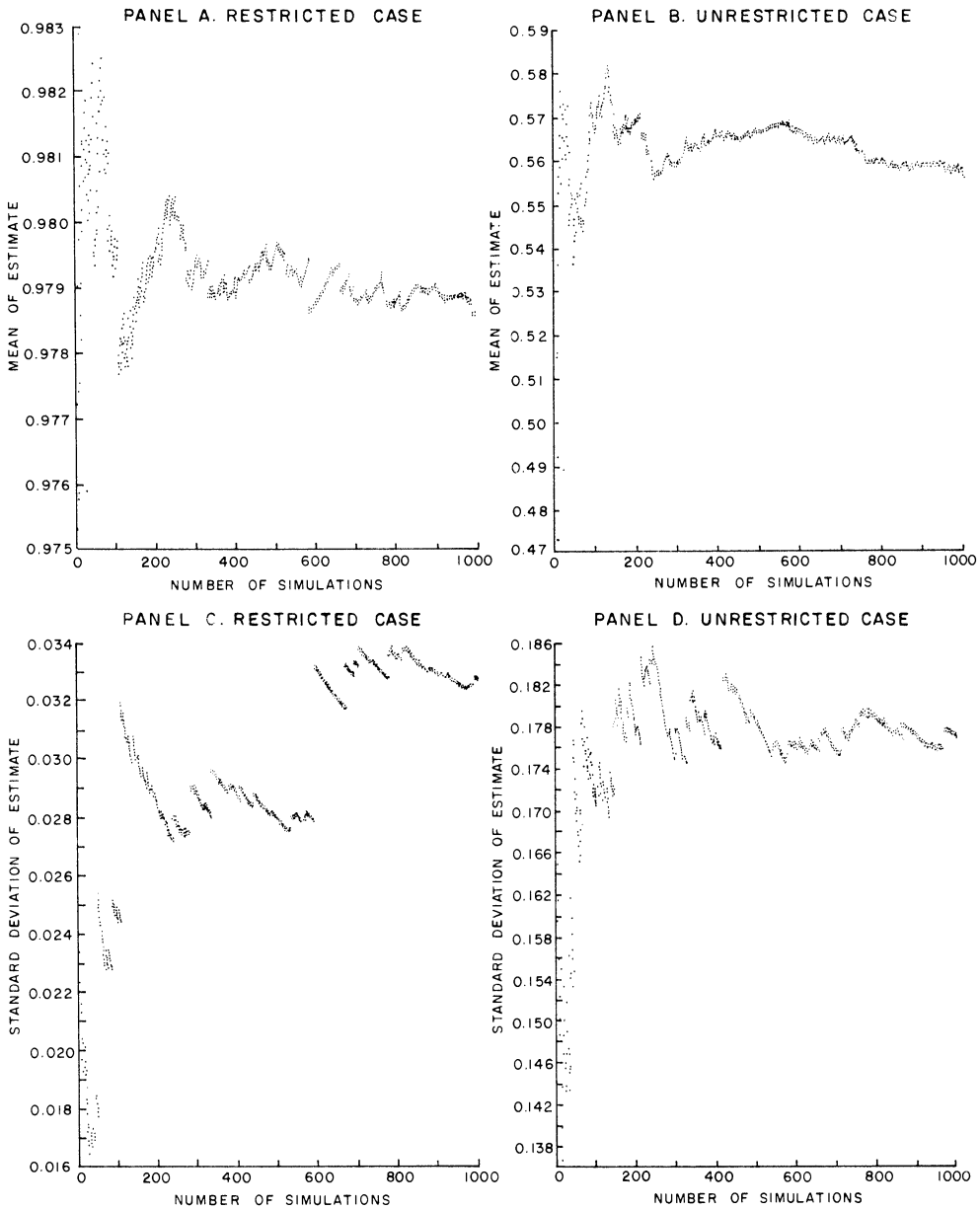


Figure B.2. Means and Standard Deviations of $\lambda = \alpha + \gamma$ Across Monte Carlo Simulations—Environment 2. Panels A and B report the sample means, and Panels C and D, report sample standard deviations across simulations as a function of the number of simulations for the restricted and unrestricted models.

Standard unit-normal variates are generated using the IMSL subroutine GGNML. The generated GARCH(1, 1) samples are estimated for the restricted model and the unrestricted (i.e., correctly specified) model. All calculations are performed in double precision FORTRAN on the IBM 3090-600E supercomputer. One thousand replications are conducted, requiring approximately 125 minutes of CPU time for each environment.

Figures B.1 and B.2 plot the means and standard deviations of the estimated λ as a function of number of replications for each environment, respectively. The convergence properties suggest that 1,000 replications are sufficient.

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